

# Kane Pavlovich

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## Professional Profile

PhD candidate in psychology / computational neuroscience with strong problem solving skills and 4+ years experience in data analytics, specializing in predictive modelling and extracting insights from complex datasets. Expertise in **predictive analysis, advanced statistics, network/signal analysis, behavioural analysis, machine learning, and data visualisation** with a proven ability to translate technical findings for diverse stakeholders. Passionate about developing high quality data-driven solutions rooted in rigorous quantitative analysis, with experience leading and communicating multiple cross-disciplinary projects from conception to implementation.

## Skills

- **Python** for machine learning (Classification & Regression) (Pandas, NumPy, scikit-learn)
- **R** for advanced statistical analyses, psychometrics (Structural Equation modelling) (dplyr, tidyverse, lavaan)
- **MATLAB** for dimensionality reduction, time-series analysis, network/graph analysis
- **Linux/Unix** for command line, Cluster Computing & parallel analyses (Slurm)
- **SQL** for database analysis, **Git** for version control, **GitHub** for publishing reproducible code.
- **Data Visualisation** with Python, R, MATLAB, and experience with Tableau.

## Education

### PHD | JANUARY 2022 - JULY 2025 | MONASH UNIVERSITY

- Developed predictive brain-behaviour models using **machine learning, network analysis, cluster analysis, psychometrics, and statistical modeling**, establishing scalable frameworks for high-dimensional neural and behavioural data.
- Published 3 first author peer-reviewed research in top academic journals ([Google Scholar](#)), with emphasis on interpretable data visualisation and reproducible open source methodology to inform best practices for future research in brain-behaviour predictive analyses.
- Presented work at international conferences (HiTOP 2025, Maths in the Brain 2023 & 2024), communicating data-driven insights to interdisciplinary stakeholders (scientists, engineers, clinicians).
- Co-supervised 2 honours students in statistical methods and engineering, leading them to produce high quality honours theses.

### MASTERS OF PSYCHOLOGY | FEBRUARY 2018 - OCTOBER 2019 | UNIVERSITY OF AUCKLAND

- Applied **time-series analysis** and **spectral analysis** (e.g., Fourier transforms, wavelet decompositions) to investigate attention-driven oscillatory dynamics in neural data, demonstrating strong **signal processing** and quantitative research skills (awarded with First Class Honours).

## Work Experience

### HEAD TUTOR / ASSISTANT LECTURER | MONASH UNIVERSITY | FEBRUARY 2025 – PRESENT

- Designed and delivered statistics & psychology focused curriculum for 500+ undergraduates, translating complex technical concepts into accessible lectures and tutorials for diverse audiences.
- Led a team of 12 tutors to standardise instruction, optimise grading workflows, and implement feedback systems, demonstrating operational leadership.

### TEACHING FELLOW | MONASH UNIVERSITY | FEBRUARY 2024 – FEBRUARY 2025

- Synthesized feedback from students, faculty, and course directors to design and optimise; lecture materials, assessments, and learning tools for psychology courses, ensuring alignment with measurable outcomes (e.g., exam performance, engagement metrics) to maximise student learning.